

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed and executed daily laboratory sample analysis priorities, ensuring timely completion of over **500 analyses** weekly to meet periodic laboratory plans and key performance indicators.
- Maintained comprehensive laboratory records and checklists to GMP standards, documenting **all analytical activities** and ensuring adherence to internal procedures and regulatory requirements.
- Performed routine calibration and maintenance for **over 15 distinct laboratory instruments**, including corrective and preventative actions, ensuring continuous operational readiness.
- Identified and escalated discrepancies, deviations, and non-conformances in testing procedures to QC management, contributing to the resolution of **approximately 8 OOS/OOT investigations** per quarter.
- Developed Python scripts to automate data aggregation for **daily quality control reporting**, reducing manual effort by an estimated 6 hours per week for the team.
- Collaborated with internal departments using **Hovione's IT platforms** to communicate QC work status and facilitate cross-functional understanding of analytical services.

SKILLS

Quality Control – GMP practices, HSE regulations, OOS/OOT investigation, analytical theory, laboratory instrument calibration, corrective maintenance, preventative maintenance, data integrity, regulatory requirements, client specifications, contract requirements

Laboratory Operations – Sample analysis, laboratory testing schedule, quality assurance standards, company procedures, daily priorities, laboratory records, laboratory notebooks, hygiene and housekeeping, instrument parts and consumables, audit support

Data Analysis & Reporting – Power BI, DAX, Power Query, Tableau, R/RStudio, pandas, NumPy, scikit-learn, statistical analysis, exploratory data analysis, time-series analysis, data modelling, data warehousing, BI dashboards

Programming & Scripting – Python, Java, Spring Boot, REST APIs, Flask, FastAPI, SQL, PostgreSQL, MySQL, Oracle, shell scripting

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins, CI/CD pipelines, code reviews

Tools & Methodologies – Jira, Confluence, Linux, Splunk, Dynatrace, Agile/Scrum ceremonies, change-control management

PROJECTS

Automated QC Data Aggregation and Reporting System (HCL Technologies)

- Developed a Python-based system to automatically aggregate and process raw data from multiple laboratory instruments, reducing manual data entry by **~75%**.
- Engineered data validation routines using **pandas and SQL** to ensure accuracy and compliance with GMP standards before generating reports.
- Created interactive **Power BI dashboards** to visualize key quality control metrics and trends for management review, facilitating faster decision-making.

Enhanced Instrument Maintenance and Calibration Tracker (MSc Project)

- Designed and implemented a relational database schema in **PostgreSQL** to track instrument calibration schedules, maintenance logs, and consumable inventory.
- Built a web interface using **Flask** to allow lab technicians to log maintenance activities and view upcoming calibration due dates, improving adherence by **~30%**.
- Integrated automated email notifications for upcoming calibrations and critical maintenance alerts, minimizing downtime risk for **over 20 laboratory instruments**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore,
India**

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)